

Andrew Gordienko

gordienko.adg@gmail.com | github.com/AndrewGordienko | LinkedIn: Andrew Gordienko

Planning & Search | Robotics Simulation | Reinforcement Learning

Seeking a Toronto role | January-April 2027 (4 months)

Education

University of Toronto

B.A., Computational Cognitive Science; Computer Science coursework

Expected June 2028

Toronto, Canada

Technical Skills

ML & Robotics: PyTorch, MuJoCo, CUDA, NumPy, Gymnasium, Transformers, LeRobot

Languages: Python, C++, Rust, JavaScript

Tooling: Git, Linux, pytest, Flask, multi-GPU training (H200), Apple MPS

Work Experience

Automata Technologies

May 2026 - Present

Research Intern, Robotics and Planning

London, UK

- Built a 3D digital twin to generate and validate laboratory configurations, then extended it into a search-based layout optimizer minimizing robot travel and other workflow costs.
- Developed a deadlock search and recovery system using CP-SAT and breadth-first search to identify unsafe workflow states and generate valid recovery paths for medical-laboratory robotics.
- Built a deterministic safety gate that verifies search-generated plans before execution, taking the system from simulation to deployment on physical robotic hardware.

OutageHub (OutageHub.ca)

Nov 2024 - Present

Founder

Hybrid

- Built a Rust-based platform aggregating Canadian utility outage data into a real-time national map and API.
- Lead partnerships and business development with organizations including the Government of Canada and Canadian Red Cross around outage intelligence and API access.

Research and Projects

TinyVLA

2026

Vision-Language-Action Model Distillation for Robotics

- Distilled a 450M-parameter SmolVLA policy to 292M parameters for local deployment, reducing bf16 footprint from 749 MB to 557 MB while achieving performance comparable to the teacher on held-out evaluations.
- Deployed the distilled policy onto physical robotic hardware and debugged the simulation-to-hardware pipeline for closed-loop manipulation.

UTMIST Virtual Creatures

Sep 2023 - Apr 2024

Lead Researcher, Reinforcement Learning and Multi-Agent Planning

- Led the creation of a Karl Sims-inspired genetic algorithm to generate MuJoCo robot morphologies and trained shared PPO locomotion policies across evolved bodies to play 2v2 soccer.
- Developed multi-agent planning experiments using Transformer-generated spatial heatmaps and Monte Carlo Tree Search for action selection.

Hudson64

2025

AlphaZero-Style Reinforcement Learning and Search

- Built an AlphaZero-style chess engine combining a learned policy-value network with Monte Carlo Tree Search, reaching approximately 1800 Elo.
- Implemented PUCT search, network-guided priors, exploration, batched neural evaluation, self-play, replay buffers and arena-based model promotion.
- Scaled training from Apple MPS to NVIDIA H200 GPUs, building high-throughput infrastructure for self-play generation and policy-value training.

Additional Experience

G&K Software LLC

Nov 2025 - Present

Head of Growth

Remote

- Lead sales and business development for a software and AI consultancy, translating customer problems into scoped technical engagements.

CODE-IT Hacks

Jun 2022 - Sep 2022

Artificial Intelligence Instructor

Toronto, Canada

- Taught AI, machine learning and programming to middle and high school students through hands-on coding projects.